Summary in Graph

Exam Summary (GO Classes GATE CS | Set Theory | Weekly Quiz 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	10 (5 + 5)
Total Marks:	15 (5 + 10)
Exam Duration:	20 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained

Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10

* Average Stats based on first-attempt stats of Top 10% (56) out of 560 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Set Theory & Algebra

What is the Cardinality of the Power set of the set $\{0, 1, 2\}$?

A. 8
B. 6
C. 7
D. 9

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

The symmetric difference of sets $\mathrm{A} = \{1, 2, 3, 4, 5, 6, 7, 8\}$ and $\mathrm{B} = \{1, 3, 5, 6, 7, 8, 9\}$ is:

A. $\{1, 3, 5, 6, 7, 8\}$
B. $\{2, 4, 9\}$
C. $\{2, 4\}$
D. $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

If $A = \{x, y, z\}$ and $B = \{u, v, w, x\}$, and the universe is $\{s, t, u, v, w, x, y, z\}$. Then $(\overline{A \cup B}) \cap (\overline{A \cap B})$ is equal to

A. $\{u, v, w, x\}$
B. $\{x\}$
C. $\{u, v, w, x, y, z\}$
D. $\{u, v, w\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

The power set of the set $\{\Phi\}$ is

A. $\{\Phi\}$
B. $\{\Phi, \{\Phi\}\}$
C. $\{0\}$
D. $\{0, \Phi, \{\Phi\}\}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 13sec | Discuss

Q #5 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Set Theory & Algebra

Let A, B be two sets. Let $\bar{A}$ denote the complement of set A (with respect to some fixed universe), and $(A - B)$ denote the set of elements in A which are not in B. Set $(A - (A - B))$ is equal to:

A. B
B. $A \cap \bar{B}$
C. $A - B$
D. $A \cap B$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Let S be an infinite set and $S_1 \ldots, S_n$ be sets such that $S_1 \cup S_2 \cup \cdots \cup S_n = S$. Then

A. at least one of the sets S_i is a finite set
B. not more than one of the sets S_i can be finite
C. at least one of the sets S_i is an infinite
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Set Theory & Algebra

Power set of empty set has exactly _____ subsets.

A. One
B. Two
C. Zero
D. Three

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Which of the following is/are true?

A. If S is a set and $|S| = 103$, then S is not the power set of any set (that is, there is no set T where $S = \mathcal{P}(T)$).
B. If S is a set and $|S| = 103$, then S is a power set of some set (that is, there is some set T where $S = \mathcal{P}(T)$).
C. If S is a set and $|S| = 8$, then S is a power set of some set (that is, there is some set T where $S = \mathcal{P}(T)$).
D. If S is a set and $|S| = 8$, then S is not the power set of any set (that is, there is no set T where $S = \mathcal{P}(T)$).

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Which of the following statements is /are TRUE?

A. $2 \in A \cup B$ implies that if $2 \notin A$ then $2 \in B$.
B. $\{2, 3\} \subseteq A$ implies that $2 \in A$ and $3 \in A$.
C. $A \cap B \supseteq \{2, 3\}$ implies that $\{2, 3\} \subseteq A$ and $\{2, 3\} \subseteq B$.
D. $A - B \supseteq \{3\}$ and $\{2\} \subseteq B$ implies that $\{2, 3\} \subseteq A \cup B$.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Set Theory & Algebra

Which of the following statements is /are False?

A. $\{2, 3, 4\} \in A$ and $\{2, 3\} \in B$ implies that $\{4\} \subseteq A - B$.
B. $A \cap B \supseteq \{2, 3, 4\}$ implies that $\{2, 3, 4\} \subseteq A$ and $\{2, 3, 4\} \subseteq B$.
C. $A - B \supseteq \{3, 4\}$ and $\{1, 2\} \subseteq B$ implies that $\{1, 2, 3, 4\} \subseteq A \cup B$.

D. $\{2,3\} \subseteq A \cup B$ implies that if $\{2,3\} \cap A = \emptyset$ then $\{2,3\} \subseteq B$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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